



$$\begin{cases} x = |r| \cos \alpha \\ y = |r| \sin \alpha \end{cases}$$

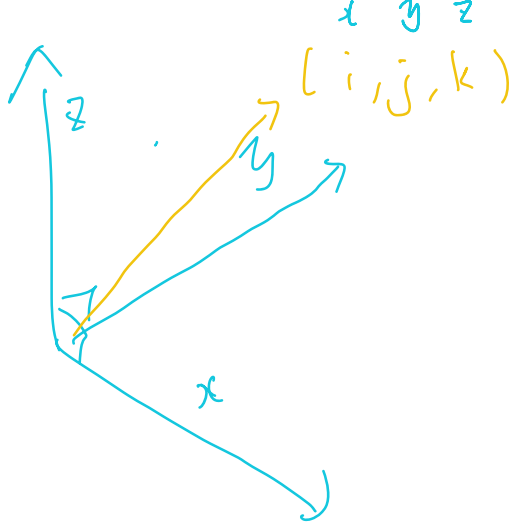
$$\begin{cases} x' = |r| \cos (\alpha + \beta) \\ y' = |r| \sin (\alpha + \beta) \end{cases}$$

$$\begin{aligned} x' &= |r| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ &= x \cos \beta - y \sin \beta \end{aligned}$$

$$\begin{aligned} y' &= |r| (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ &= x \sin \beta + y \cos \beta \end{aligned}$$

∴ The rotational matrix with an angle θ on a 2-D plane.

$$\therefore \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



Anti-clockwise.
rotation along the z -axis,
i.e., on the x - y plane.

$$\begin{cases} x = |r_z| (\cos \alpha) \\ y = |r_z| (\sin \alpha) \\ z = k \end{cases}$$

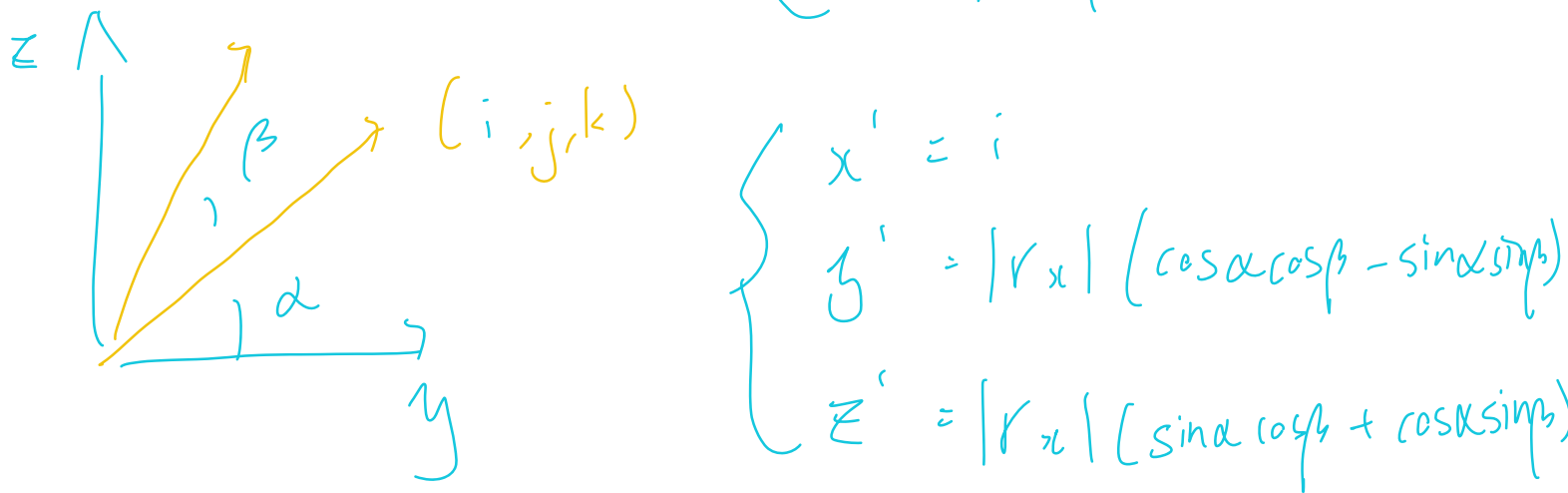
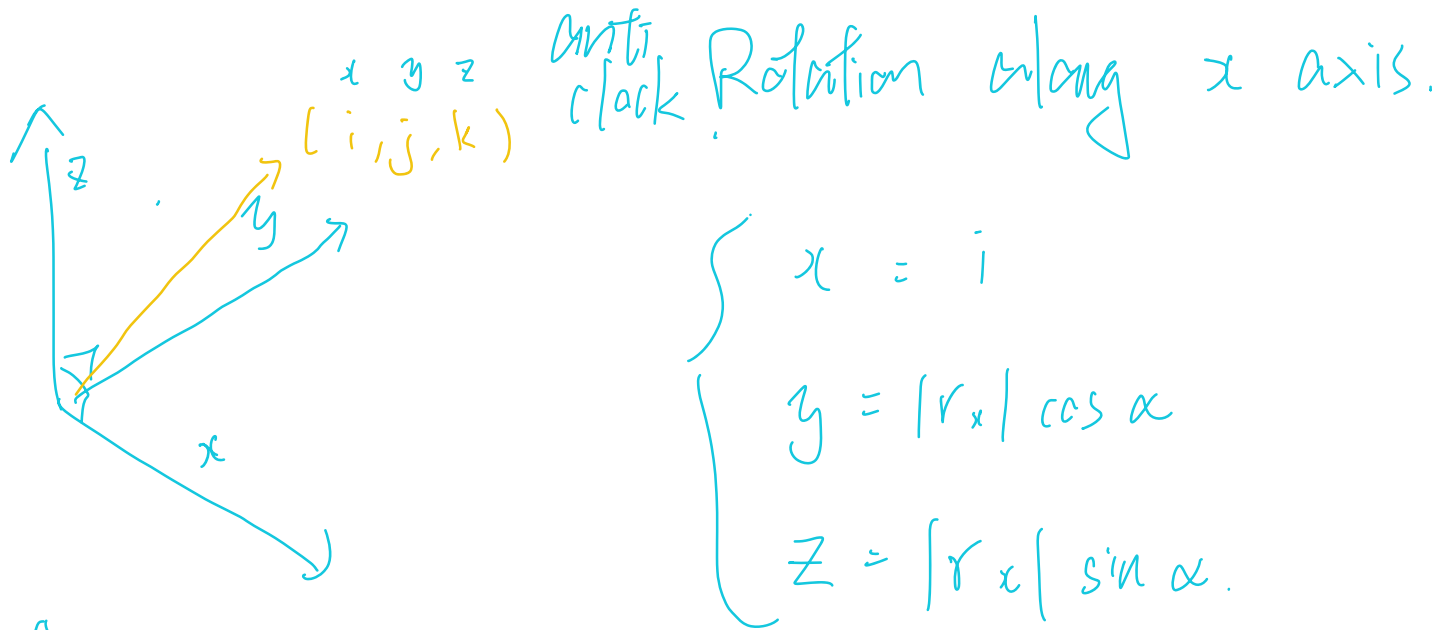


$$\begin{cases} x' = |r_z| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \\ y' = |r_z| (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ z = k \end{cases}$$

$$\therefore \begin{cases} x' = x \cos \beta - y \sin \beta + z (0) \\ y' = x \sin \beta + y \cos \beta + z (0) \\ z' = z (1) + z (0) + z (0) \end{cases}$$

$\therefore$ the rotational matrix around z axis

$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



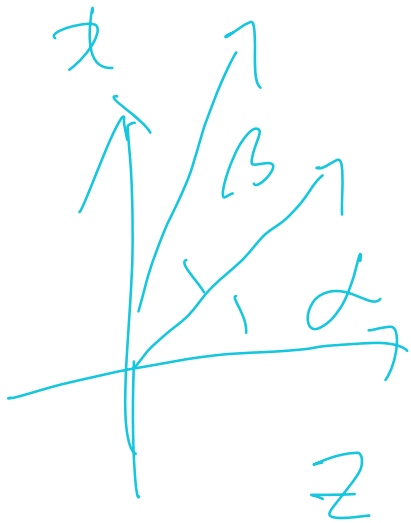
$$\therefore \begin{cases} x' = x \\ y' = y \cos \beta - z \sin \beta \\ z' = y \sin \beta + z \cos \beta \end{cases}$$

$$\therefore \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta \\ 0 & \sin \beta & \cos \beta \end{bmatrix}$$

ortho-dock. Along y axis.

$$\begin{cases} x = |r_y| \sin \alpha \\ y = j \\ z = |r_y| \cos \alpha. \end{cases}$$

$$\begin{cases} x' = |r_y| (\sin \alpha \cos \beta + \cos \alpha \sin \beta) \\ y' = j \\ z' = |r_y| (\cos \alpha \cos \beta - \sin \alpha \sin \beta) \end{cases}$$



$$\begin{cases} x' = x \cos \beta + z \sin \beta \\ y' = j \\ z' = -x \sin \beta + z \cos \beta. \end{cases}$$

$$\therefore \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$